



ECEN 314—Signals and Systems

Scott L. Miller

Module 22

Application: Analog Communications
Part 2 – Phase/Frequency Modulation

Learning Objectives



- Understand how phase and frequency modulation work.
- Be able to construct circuitry to function as AM/FM modulators and demodulators.
- Be able to compute the Carson Rule BW of an FM signal.
- Understand the basics of how a PLL works and what they are used for.

Phase/Frequency Modulation



In amplitude modulation, the message was related to the amplitude of the carrier and the phase was constant

$$s(t) = A(t)\cos(\omega_c t + \theta)$$

$$\text{AM: } A(t) = A + \mu \cdot m(t),$$

$$\text{DSBsc: } A(t) = \mu \cdot m(t).$$

In angle modulation, the message is somehow related to the phase of the carrier and the amplitude is constant.

$$s(t) = A\cos(\omega_c t + \theta(t))$$

Phase modulation (PM):

$$\theta(t) = k_p m(t),$$

Frequency modulation (FM):

$$\frac{1}{2\pi} \frac{d\theta}{dt} = k_f m(t).$$

Some terminology

✓ Instantaneous phase:

$$\psi_i(t) = \omega_c t + \theta(t),$$

✓ Phase deviation: $\theta(t)$,

✓ Phase deviation constant: k_p ,

✓ Peak phase deviation:

$$\Delta\theta = \max_t \theta(t) = k_p \cdot \max_t m(t),$$

✓ Phase modulation index:

$$\mu_p = \Delta\theta.$$

✓ Instantaneous frequency:

$$f_i(t) = \frac{1}{2\pi} \frac{d\psi_i}{dt} = f_c + \frac{1}{2\pi} \frac{d\theta}{dt},$$

✓ Frequency deviation: $\frac{1}{2\pi} \frac{d\theta}{dt}$,

✓ Frequency Deviation constant: k_f ,

✓ Peak frequency deviation:

$$\Delta f = \max_t \frac{1}{2\pi} \frac{d\theta}{dt} = k_f \cdot \max_t m(t),$$

✓ Frequency modulation index:

$$\mu_p = \Delta f / B \text{ (where } B = \text{BW of } m(t)\text{)}.$$

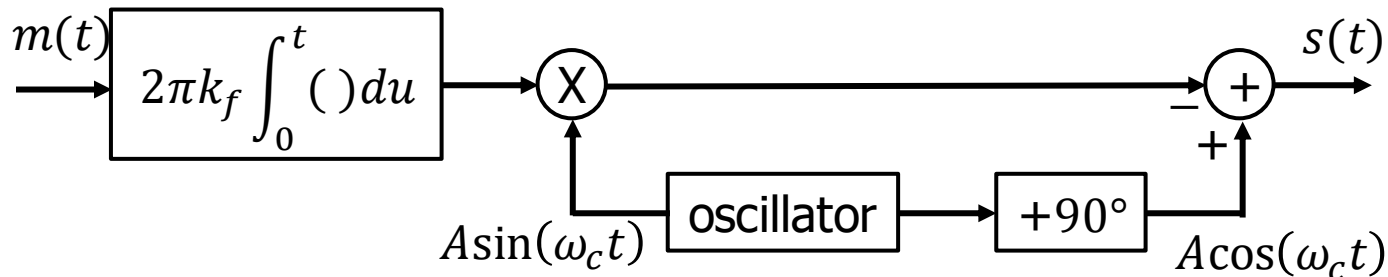
Narrowband FM Modulator



Consider the case of FM (or PM) where the modulation index is small enough so that $\theta(t) \ll 1$. Then the transmitted signal can be written as

$$\begin{aligned} s(t) &= A \cos(\omega_c t + \theta(t)) = A \cos(\omega_c t) \cos(\theta(t)) - A \sin(\omega_c t) \sin(\theta(t)) \\ &\approx A \cos(\omega_c t) - A \theta(t) \sin(\omega_c t) \end{aligned}$$

For the case of FM, $\theta(t) = 2\pi k_f \int_0^t m(u) du$, and the resulting signal is known as narrowband FM.

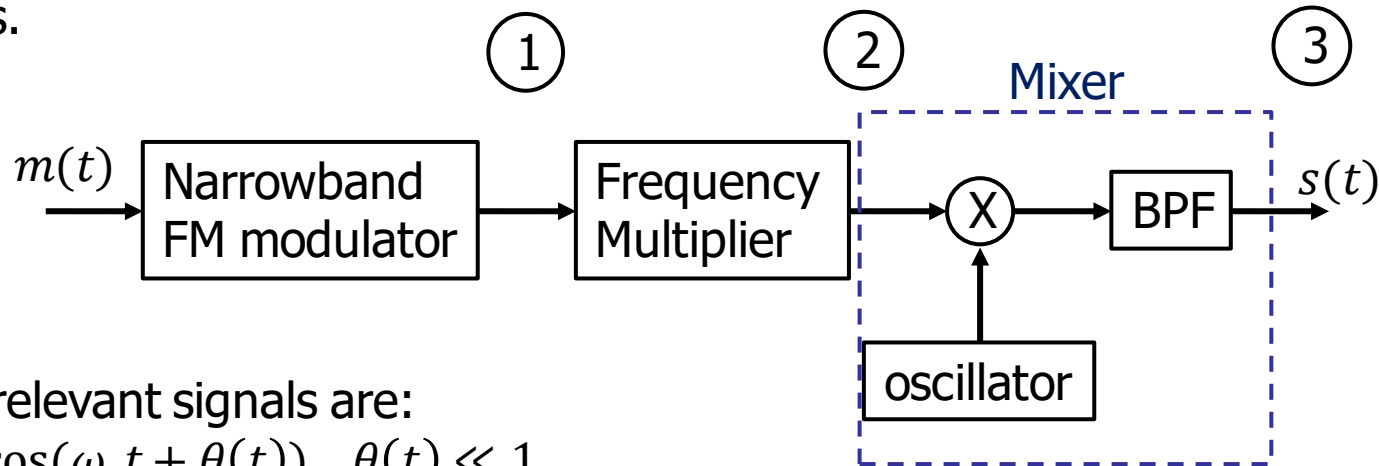


Note: We know how to build/implement all of the blocks in the block diagram from previous discussions in this class or from previous classes (e.g., 214).

Wideband FM Modulator



Once a narrowband FM signal is generated, it can be converted to wideband FM as follows.

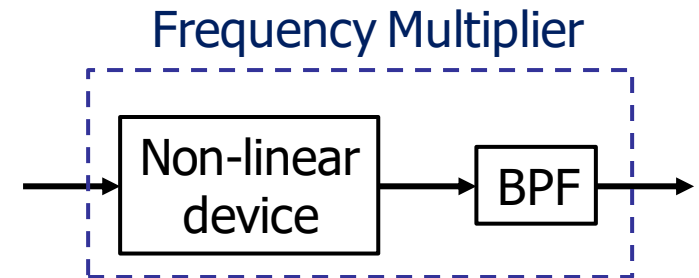


The relevant signals are:

- 1) $\cos(\omega_c t + \theta(t))$, $\theta(t) \ll 1$,
- 2) $\cos(n\omega_c t + n\theta(t))$,
- 3) $\cos(\omega_c t + n\theta(t))$.

The frequency multiplier can be implemented by a non-linear device followed by a BPF.

The non-linear device generates all harmonics of the input frequency and the BPF filters the desired harmonic.



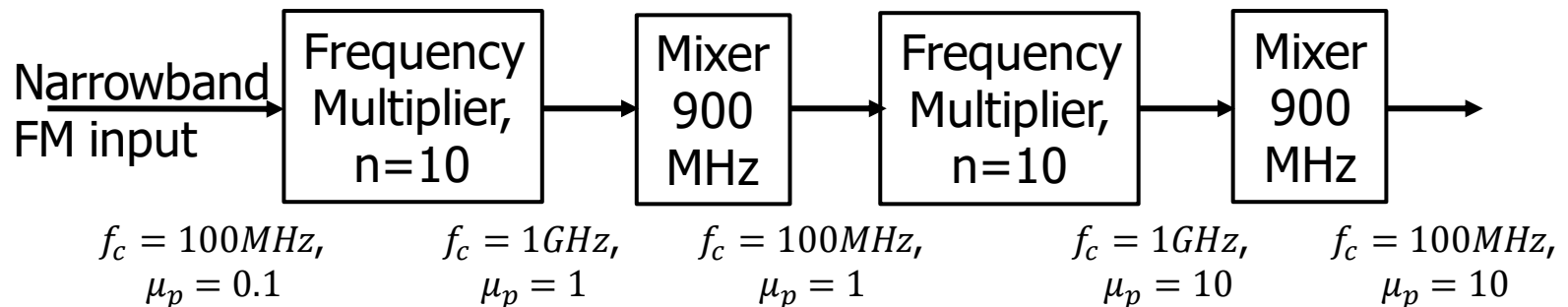
Example



Suppose we generate a narrowband PM signal with $f_c = 100\text{MHz}$ and $\mu_p = 0.1$. We wish to convert this to $f_c = 100\text{MHz}$ and $\mu_p = 10$. This will require:

- ✓ frequency multiplication factor, $n = 100$
- ✓ Mixer's oscillator frequency = 9.9GHz or 10.1GHz.

If it is inconvenient to have such a high oscillator frequency, the narrowband-to-wideband conversion can be done in stages.



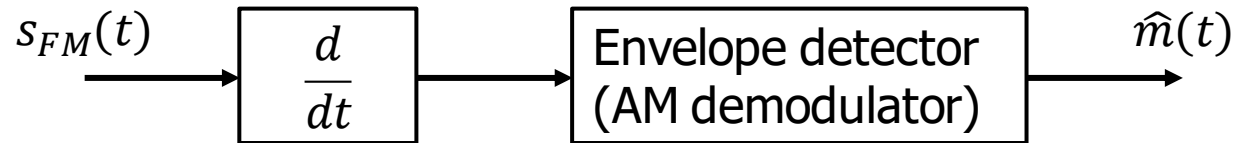
Or we could do it in 3 stages. Maybe something like:

- ✓ Stage 1 – $n = 5$, $f_o = 400\text{MHz}$, (or 600MHz) $\rightarrow \mu_p = 0.5$
- ✓ Stage 2 – $n = 5$, $f_o = 400\text{MHz}$, (or 600MHz) $\rightarrow \mu_p = 2.5$
- ✓ Stage 3 – $n = 4$, $f_o = 300\text{MHz}$, (or 500MHz) $\rightarrow \mu_p = 10$

FM Demodulation



A simple method for demodulating FM signals is called the *frequency discriminator* shown below.



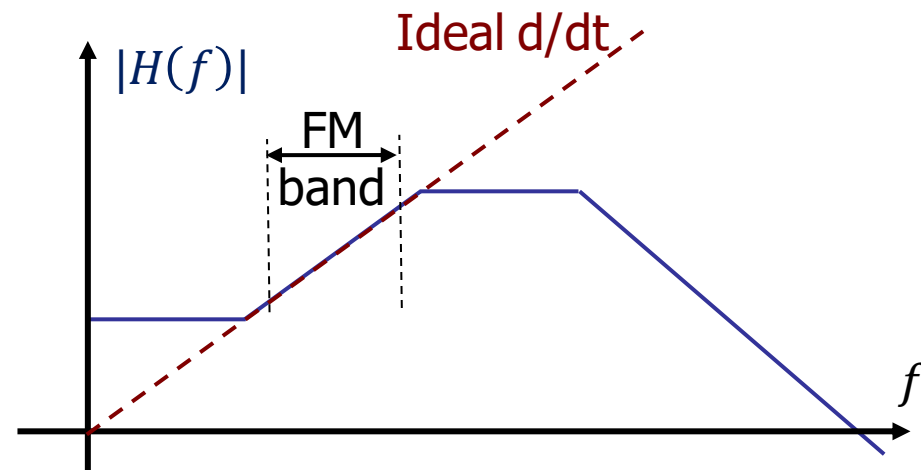
If $s_{FM}(t) = A \cos(\omega_c t + 2\pi k_f \int_0^t m(u) du)$, then

$$\frac{ds_{FM}}{dt} = A(\omega_c + 2\pi k_f m(t)) \cos(\omega_c t + 2\pi k_f \int_0^t m(u) du),$$

$$\hat{m}(t) = A[\omega_c + 2\pi k_f m(t)].$$

After removing the DC bias, $\hat{m}(t) \propto m(t)$.

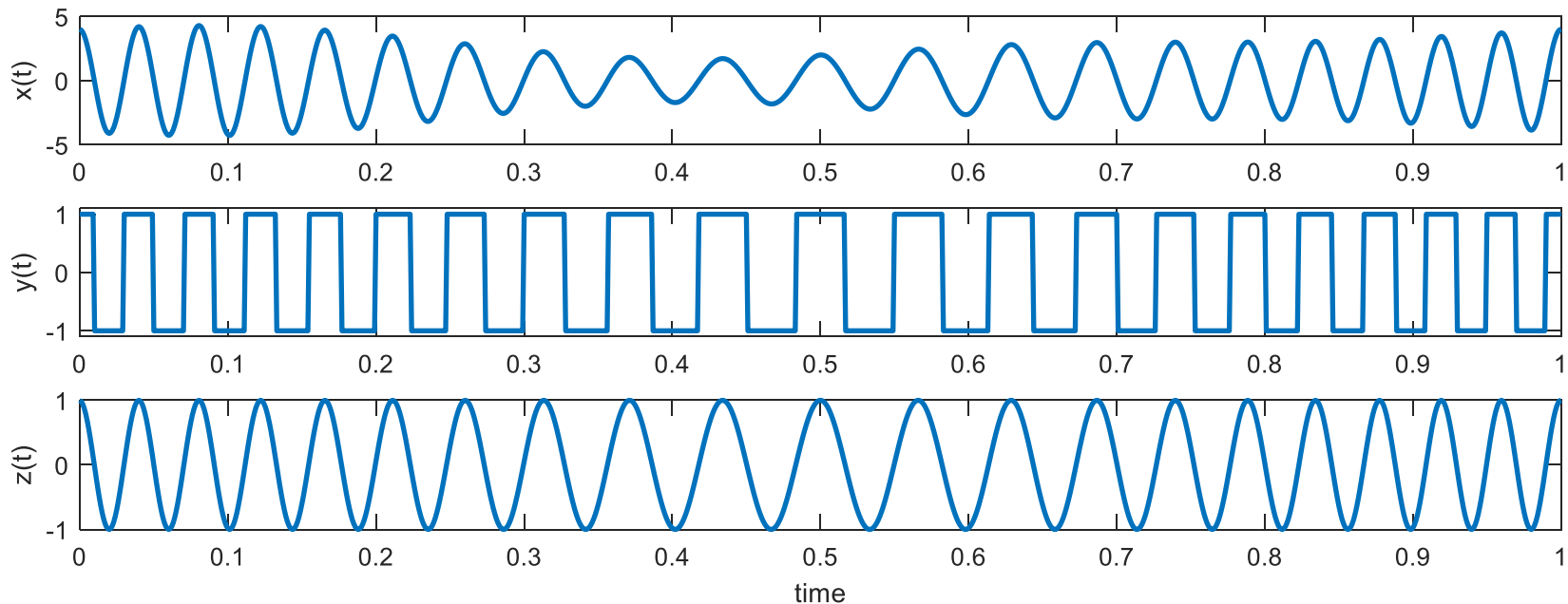
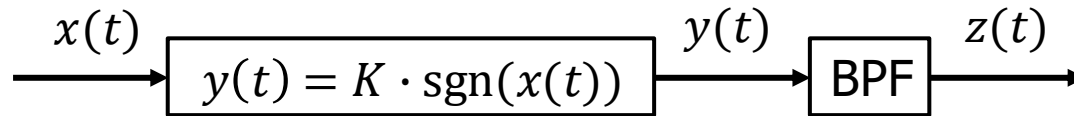
Note: The frequency discriminator calls for the use of a differentiator. A true differentiator tends to be unstable. Fortunately, we can implement a more well-behaved filter that acts like a differentiator over the width of the FM band.



Bandpass Limiter



The frequency discriminator essentially converts the frequency deviations to amplitude deviations and then uses a AM demodulator to detect the signal. With such an approach, it is necessary to make sure the received FM signal does not have any unwanted amplitude modulation (perhaps due to fluctuations in received signal strength over time). This is achieved with a device known as a *bandpass limiter*.



Voltage Controlled Oscillator



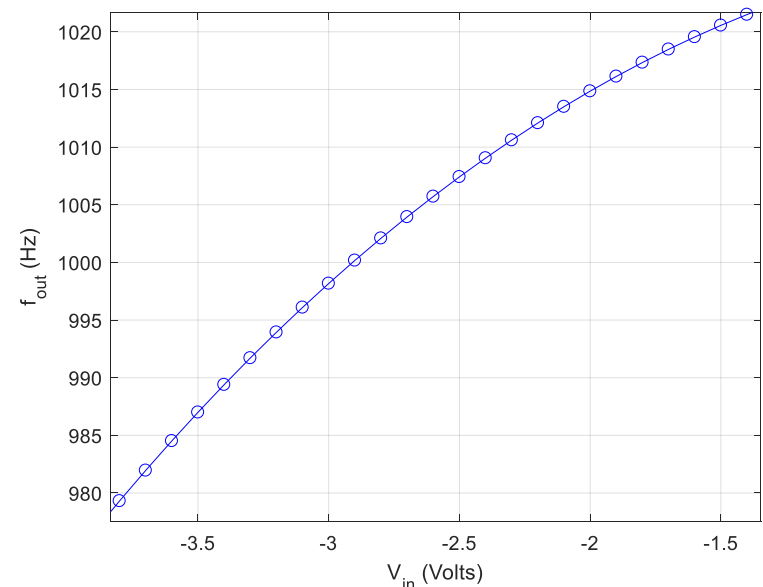
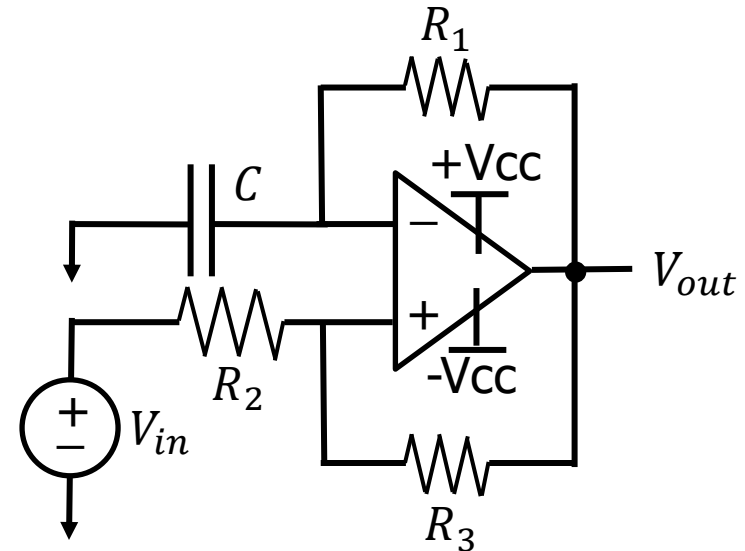
A more direct approach to creating an FM signal would be to use a circuit that functions as a *voltage controlled oscillator* (VCO). Here is one based off of a circuit you studied in ECEN 214 (Lab 6).

The output should be a square wave whose frequency depends on the input voltage.

You could follow this up with a LPF to extract the fundamental frequency if you want a sinusoidal carrier.

The plot shows the output frequency as a function of input voltage for:

- ✓ $R_1 = 10k\Omega$,
- ✓ $R_2 = 30k\Omega$,
- ✓ $R_3 = 10k\Omega$,
- ✓ $C = 25nF$,
- ✓ $V_{CC} = \pm 9V$.



Spectrum of an FM Signal



It is not possible to evaluate (in closed form) the spectrum of a PM/FM signal in general, so we consider a specific example in order to gain some insight. Let $\theta(t) = \mu \sin(\omega_m t)$.

This could result from:

- ✓ A PM signal with $m(t) \propto \sin(\omega_m t)$ and $\mu_p = \mu$, or
- ✓ An FM signal with $m(t) \propto \frac{d\theta}{dt} = \mu \omega_m \cos(\omega_m t)$ and $\mu_f = \mu$.

Rewrite the angle modulated signal as

$$s(t) = A \cos(\omega_c t + \mu \sin(\omega_m t)) = \text{Re}[e^{j\omega_c t} e^{j\mu \sin(\omega_m t)}].$$

Note that $x(t) = e^{j\mu \sin(\omega_m t)}$ is a periodic function with period of $T_o = \frac{1}{f_m}$ and therefore it has a Fourier Series representation

$$e^{j\mu \sin(\omega_m t)} = \sum_{k=-\infty}^{\infty} X_k e^{jk\omega_m t}.$$

Furthermore, since $x(-t) = x^*(t)$, the X_k are purely real (no imaginary part).

Spectrum of an FM Signal

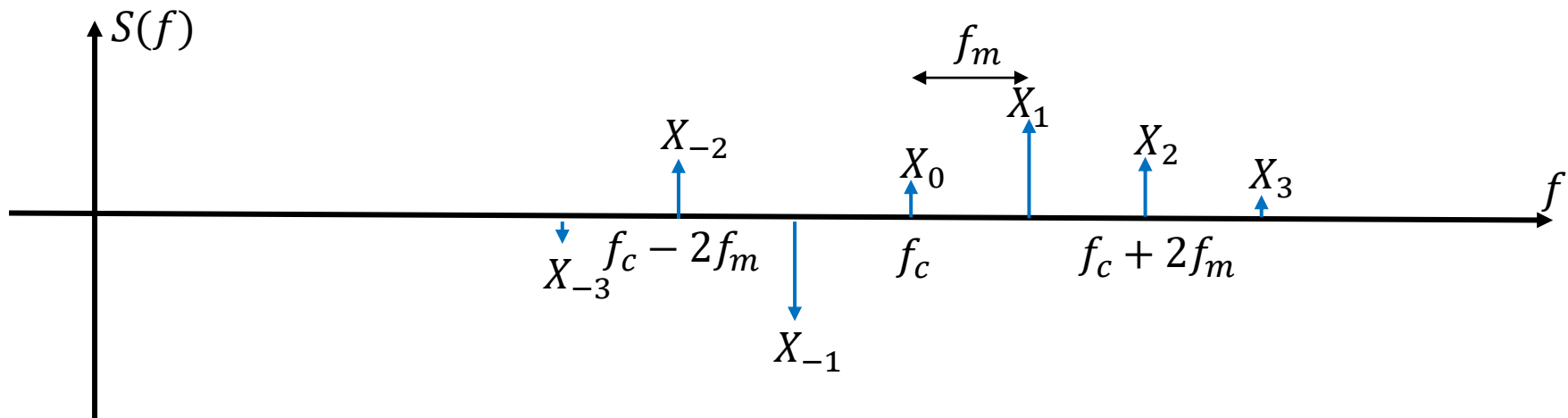


Next write $s(t)$ as

$$\begin{aligned} s(t) &= \text{Re}[Ae^{j\omega_c t} e^{j\mu \sin(\omega_m t)}] = \text{Re}[Ae^{j\omega_c t} \sum_{k=-\infty}^{\infty} X_k e^{jk\omega_m t}] \\ &= A \sum_{k=-\infty}^{\infty} X_k \text{Re}[e^{j(\omega_c + k\omega_m)t}] = A \sum_{k=-\infty}^{\infty} X_k \cos((\omega_c + k\omega_m)t). \end{aligned}$$

The spectrum of $s(t)$ is then found to be

$$S(f) = \mathcal{F}\{s(t)\} = \frac{A}{2} \sum_{k=-\infty}^{\infty} X_k [\delta(f - (f_c + kf_m)) + \delta(f - (f_c - kf_m))].$$



Spectrum of an FM Signal



Finally, we need to find the values of the F.S. coefficients

$$X_k = \frac{1}{T_o} \int_0^{T_o} e^{j\mu \sin(\omega_m t)} e^{-jk\omega_m t} dt.$$

This integral cannot be evaluated in closed form, but it can be written in terms of well-known (tabulated) functions using the change of variables $\varphi = \omega_m t$.

$$X_k = \frac{1}{2\pi} \int_0^{2\pi} e^{j[\mu \sin(\varphi) - k\varphi]} d\varphi = J_k(\mu).$$

The F.S. coefficients can be evaluated using the k^{th} order Bessel functions ("bessej(k,mu)" in MATLAB)

Example

For a modulation index of $\mu = 2$ the F.S. coefficients are as shown in the table. Note that most of the power in the signal is in a band within $\pm 3f_m$ of f_c .

k	$X_k = J_k(2)$	% of total power within ($f_c - kf_m, f_c + kf_m$)
0	0.2239	5.013
1	0.5767	71.54
2	0.3528	96.43
3	0.1289	99.76
4	0.0340	99.99
5	0.0070	100.00

Spectrum of an FM Signal



Expanding on the previous example, the plot shows the % of Power vs. BW for several different values of the modulation index when the message signal is sinusoidal.

The dashed line is the 98% power level and the circles show where the curves intersect the 98% level.

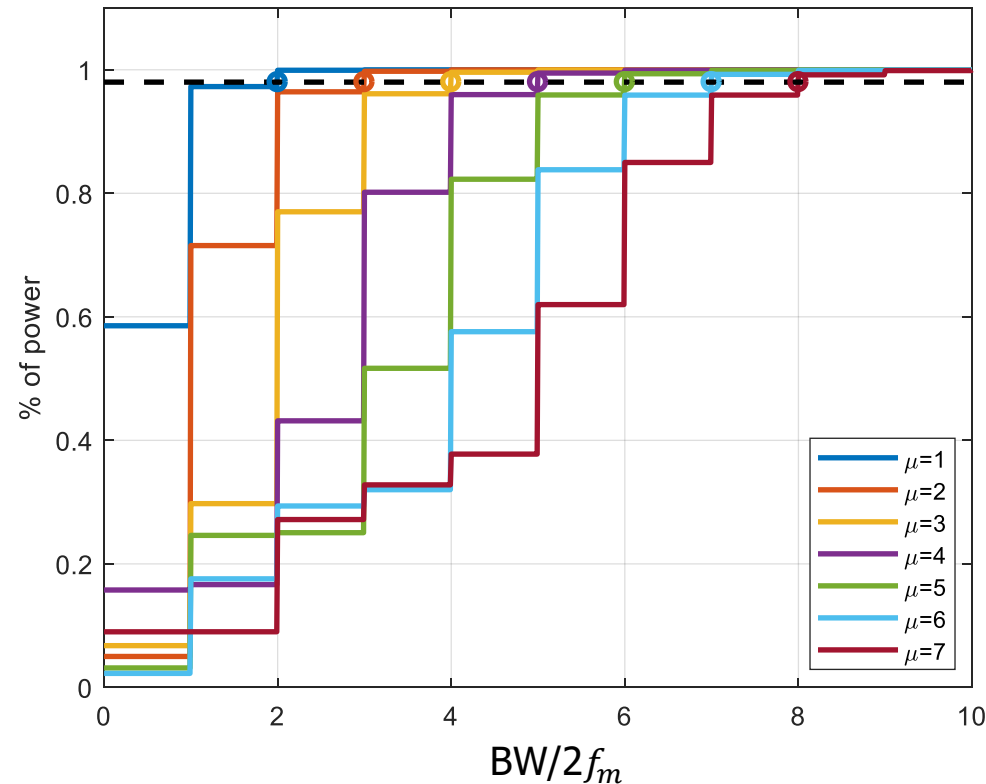
From this plot we see that for sinusoidal messages the 98% power BW is given by

$$B_{98\%} = 2(\mu + 1)f_m.$$

Generalizing this result to other types of messages produces what is known as the *Carson Rule* BW for FM signals:

$$BW = 2(\mu + 1)B$$

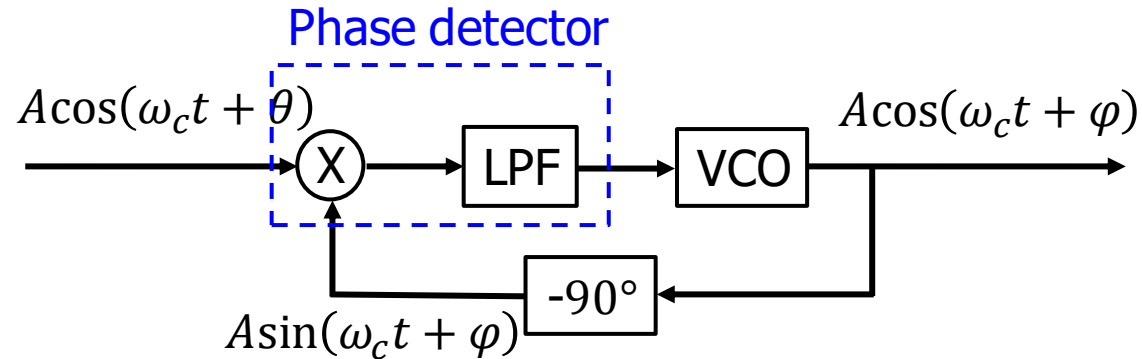
where B is the BW of the message signal.



Phase Locked Loop (PLL)



A Phase lock loop (PLL) is a feedback system used to produce a sinusoidal signal that is locked in frequency/phase to an input signal.



Assuming the VCO is locked in frequency but not necessarily in phase, the output of the multiplier is

$$A^2 \cos(\omega_c t + \theta) \sin(\omega_c t + \varphi) = \frac{A^2}{2} [\sin(2\omega_c t + \varphi + \theta) + \sin(\varphi - \theta)].$$

The LPF will remove the double frequency term so that the input to the VCO is $\propto \sin(\varphi - \theta)$.

The instantaneous output frequency of the VCO is proportional to its input voltage

$$\omega_c + \frac{d\varphi}{dt} = \alpha + \beta \sin(\varphi - \theta).$$

If we set up the VCO so that its free-running frequency is $\alpha = \omega_c$, and $\beta > 0$, then

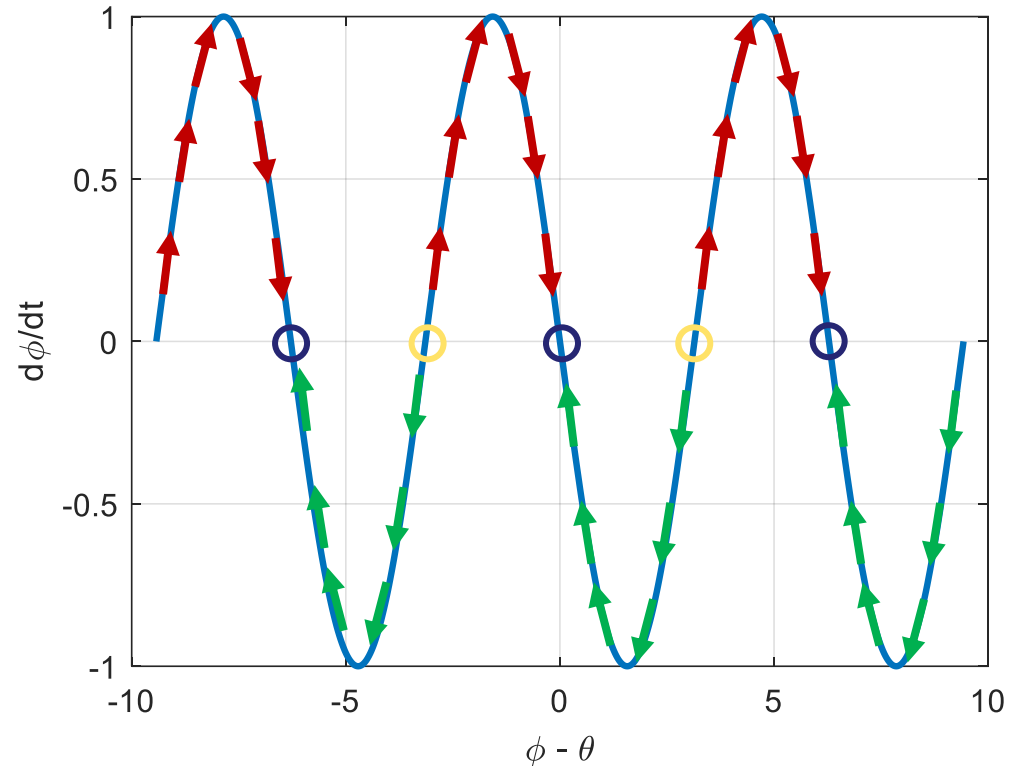
$$\frac{d\varphi}{dt} \propto \sin(\varphi - \theta).$$

Phase Locked Loop (PLL)



This plot shows how this feedback system changes the phase of the oscillator as a function of the difference between the phase of the local oscillator and the phase of the incoming signal.

This plot is known as a phase plane plot and helps us to understand the stable points of the PLL.



Case I (red arrows): When $\sin(\phi - \theta) > 0$, $\frac{d\phi}{dt} > 0$, $\rightarrow \phi - \theta$ increases.

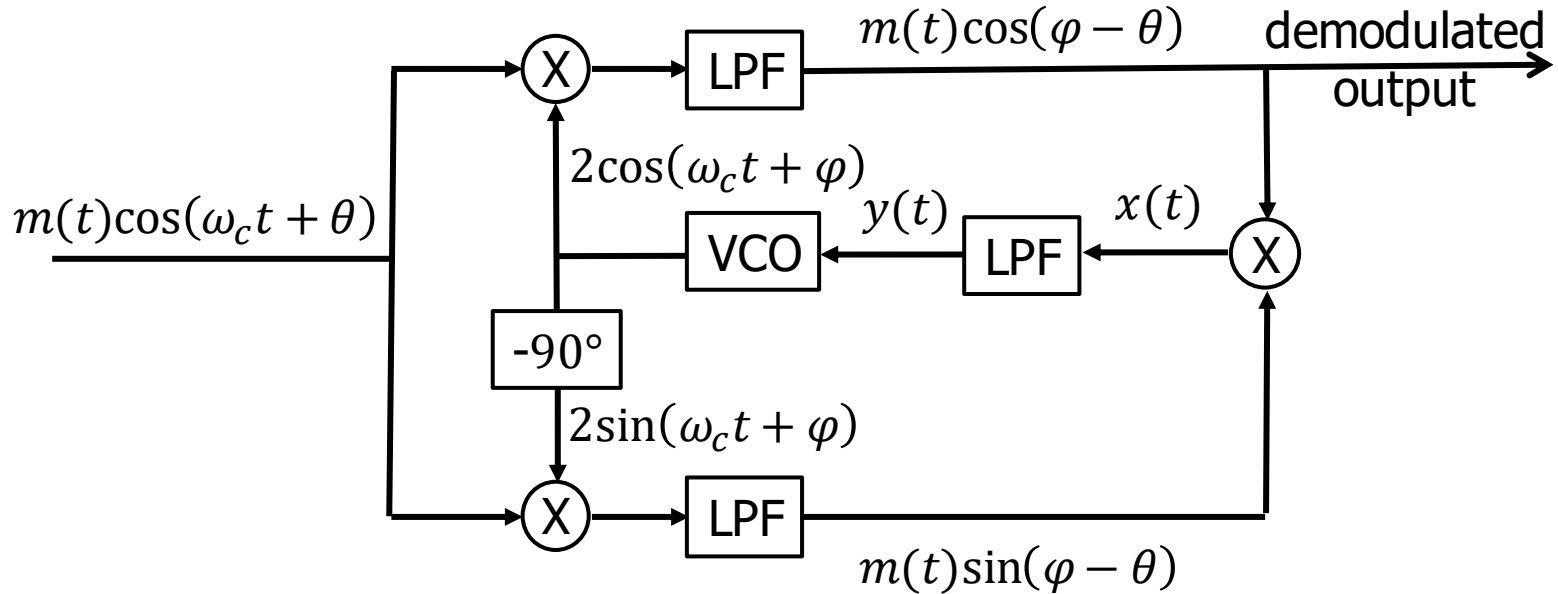
Case II (green arrows): When $\sin(\phi - \theta) < 0$, $\frac{d\phi}{dt} < 0$, $\rightarrow \phi - \theta$ decreases.

The system has stable equilibrium points (blue circles) at $\phi - \theta = 2\pi n$ for any integer n . The equilibrium points at $2(n + 1)\pi$ are unstable.

Costas Loop



If the incoming signal is modulated (e.g., DSB), a modified version of the PLL known as a Costas loop is used.



$$x(t) = \frac{1}{2}m^2(t)\sin(2(\phi - \theta))$$

$$y(t) = \frac{1}{2}m^2(t)\sin(2(\phi - \theta)) \propto \sin(2(\phi - \theta))$$

Notes:

- 1) Now the stable equilibrium points will be at $\phi - \theta = n\pi$ with unstable equilibrium points at $\phi - \theta = (n + \frac{1}{2})\pi$.
- 2) At the stable equilibrium points, the output of the upper LPF will be $\propto \pm m(t)$ and so this circuit can function as a DSB demodulator.